

Slide.Bingo: From Passive Attendance to Active Listening Through AI-Generated Bingo

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Abstract

Conference attendance is predominantly passive: audiences sit, listen, and hope to absorb information. Yet, active learning research demonstrates that participation outperforms passive reception. We present Slide.Bingo, an interactive system that uses generative AI to transform presentation content into personalised bingo cards, aiming to give each attendee a task to perform and specific targets to attend to during sessions. The bingo format provides sustained

engagement, directed attention, and social presence through real-time awareness of other players. Designed for conference-scale deployment, the system supports two card policies: an *Individual Card Policy* for paper/panel sessions and a *Shared Card Policy* for demo/poster sessions. Presenters can customize their content and add custom statements for key takeaways. At CHI 2026, attendees can use Slide.Bingo as their conference schedule app, experiencing Slide.Bingo across the programme. Visitors to our interactivity booth will be able to interact with a demo presentation or customize Slide.Bingo content for their own papers.



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CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; *Collaborative and social computing systems and tools*; •

Applied computing → **Interactive learning environments**; •
Computing methodologies → Natural language processing.

Keywords

Active Learning, Gamification, Audience Engagement, Conferences, Presentations, Generative AI, Real-Time Interaction, Bingo

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1 Introduction

1.1 The Challenge of Passive Attendance

Academic conferences present a paradox: attendees travel significant distances and invest considerable time, yet the dominant mode of participation remains passive. Audiences sit, listen, and hope to absorb information, but the format offers few opportunities to support active engagement. Research on attention and learning consistently shows that passive reception is less effective than active participation [4, 5, 8], yet traditional presentation formats persist largely unchanged. The challenge is amplified at scale. A major conference like CHI packs hundreds of presentations into a few days, requiring attendees to navigate parallel sessions, switch contexts repeatedly, and maintain focus through back-to-back talks. Cognitive fatigue accumulates. Attention wanders. Conferences also encompass diverse session formats with fundamentally different participation patterns. Paper sessions involve sequential presentations to seated (sometimes immediately sleepy [7]) audiences. Demo, poster, and interactivity sessions involve concurrent exhibits where attendees move freely. These formats demand different approaches to supporting engagement.

1.2 Active Learning and Directed Attention

Educational research offers a clear prescription: active learning outperforms passive instruction. Freeman et al.'s meta-analysis of 225 studies found that active learning increases examination performance by half a standard deviation and reduces failure rates by 55% compared to traditional lecturing [5]. The core insight is simple: learners benefit from *doing something* rather than merely receiving information [2, 9]. Beyond activity alone, learning research suggests that engagement is especially effective when learners are prompted to externalize their understanding by producing brief artifacts or representations, rather than only performing reactive actions, as such constructive engagement supports deeper processing than passive reception or activity alone.

Having specific targets keeps attention focused and creates opportunities for material to be processed rather than passively heard. This aligns with Ausubel's concept of *advance organisers*: frameworks of key concepts that support learners know where to focus

their attention [1]. Gamification research suggests that game elements such as points, achievements, and clear goals can sustain participation, engagement, and motivation [10].

1.3 Existing Approaches and Their Limitations

Several technologies address audience engagement during presentations, but each has limitations. Audience response systems (clickers, live polls) enable periodic check-ins but do not scaffold sustained attention throughout a talk. Backchannel tools (Social media hashtags, dedicated chat) enable social interaction but may compete with rather than complement attention to the speaker. Conference apps provide schedules and networking but offer no structured support for the presentation experience itself.

Bingo has been used previously to engage conference attendees. CHI Play Bingo [6] demonstrated manual bingo gameplay at CHI Play 2024, using pre-authored cards with broadly applicable statements such as "Someone mentions their sample size." This demonstrated the audience's appetite for playful engagement during academic presentations. However, there is an opportunity now to provide attendees with specific listening objectives tied to presentation content, in a tool that adapts to distinct session formats and leverages AI to scale content generation across an entire conference programme.

1.4 Slide.Bingo: Gamified Engagement Through Directed Attention

This paper presents Slide.Bingo, a system that uses AI-generated bingo cards to support active engagement and, in turn, active learning during academic presentations. Each attendee receives a personalised card containing statements drawn from the session's content that is available in the conference program (and any additional content customized by presenters). The task is simple: listen during a talk (or alternatively, explore the text during exhibition sessions) and mark statements as encountered. Completing patterns (a line or a full-house) earns achievements visible to all players.

The bingo format supports engagement through several mechanisms that are designed not only to sustain attention, but to encourage active and, at key moments, constructive engagement by prompting attendees to notice, select, and occasionally externalize elements of the presented material as it unfolds:

- **A task to perform:** Rather than passively sitting, attendees are given something active to do. Scanning the card, attending to content, and marking squares is intended to maintain involvement throughout the session.
- **Directed attention:** The statements on each card provide specific targets to listen for, scaffolding attention to presentation content. Different attendees attend to different (but overlapping) subsets of the material.
- **Sustained engagement:** Unlike periodic polls, bingo gameplay is continuous; the card remains relevant throughout the talk, offering ongoing structure for attention.
- **Social presence:** Real-time features (player counts, achievement notifications, leaderboards) create awareness of fellow participants, transforming individual attendance into a shared activity.

Designed for conference-scale deployment, Slide.Bingo imports programme data to generate bingo content across hundreds of sessions. The system supports cards for individual presentations within paper sessions and for exhibition sessions (described in the next section). Presenters can optionally customize their paper’s associated content by adding statements that highlight key takeaways or provoke thought.

Slide.Bingo differs from prior work in three key respects: (1) AI-generated content enables scaling to hundreds of sessions without manual authoring; (2) personalised cards mean each attendee has unique listening targets rather than identical shared cards; and (3) content is derived from specific presentation abstracts rather than broadly-applicable conference tropes, directing attention to actual paper content.

A potential concern is that bingo gameplay itself might distract from presentation content rather than enhance attention to it. We address this through several design choices. First, card sizes are deliberately smaller than traditional bingo (3×3 or 4×3 rather than 5×5), reducing the cognitive load of scanning for matches. Second, statements are designed as quickly recognisable phrases rather than complex propositions, minimising interpretation effort. Third, marking a bingo card requires only a single tap rather than typed input. The interaction model is closer to directed note-taking than active composition. Whether this achieves the intended balance between engagement and attention is an empirical question we aim to investigate through the CHI 2026 deployment.

2 System Description

Slide.Bingo supports multiple deployment contexts (individual presentations, semester-long courses, multi-day conferences). For CHI 2026, we focus on **Multi-day Conference Mode**, an event structured hierarchically where the *Event* contains multiple *Sessions*, each configured with a *card policy* appropriate to its format. Attendees browse the conference schedule, select a session, and join directly. Content is generated from imported programme data, with optional presenter customization. An event-wide leaderboard tracks engagement across all sessions.

2.1 Card Policies

A key architectural feature is support for different *card policies* that adapt the system to different session formats.

2.1.1 Individual Card Policy (Paper & Panel Sessions). For traditional paper sessions with sequential presentations, the Individual Card Policy generates a separate bingo card for each talk. Content is drawn from the title and abstract for each talk (optionally refined with uploaded slides or manual input by the presenter). Cards use a standard grid (3×3 or 4×3, or 5×5 - scaled to the needs of the session) and automatically advance when the session moves to the next talk. Attendees join once and play continuously across all talks.

2.1.2 Shared Card Policy (Demo/Poster/Interactivity Sessions). For concurrent exhibition sessions, the Shared Card Policy generates a single card covering the entire session. The system first synthesises an *executive summary* across all titles and abstracts, then derives concepts and statements from this synthesis. Cards are typically

larger (4×4 or 5×5) and remain valid across multiple timeslots if exhibits repeat. This is intended to transform bingo from a seated listening activity into an active exploration.

2.2 AI-Driven Content Generation

Slide.Bingo generates statements that are simultaneously *salient*, *recognisable*, and *balanced* through a multi-stage pipeline powered by the Anthropic Claude API (see Figure 1).

In conference mode, content is imported from the programme: titles and abstracts provide raw material available in the public domain. For the generation of the bingo card statements, we use the paid Anthropic Claude API — ensuring that data (i.e. titles and abstracts) are not retained for further AI training. The system extracts approximately 20 key concepts per talk, each annotated with probability (high, medium, low) and timing metadata. For each concept, multiple statement variations are generated (typically four), yielding approximately 80 candidate statements per talk. Statements are crafted as concise phrases (under 10 words) using active voice, describing observable content rather than interpretations.

2.3 Presenter Customization and Agency

A core design principle is that presenters retain meaningful control over how their work is represented. Automation should augment rather than override the presenter’s intent. By default, we will include the entire CHI2026 programme, with AI-generated content derived from titles and abstracts. Presenters are not required to take action to participate. However, we provide two options to respect presenter preferences:

- **Customization:** Presenters can visit our booth to add custom statements highlighting key takeaways or provocations. These are blended with AI-generated statements in the card pool, increasing the likelihood that presenter-chosen content appears on audience cards. Alternatively, they can provide additional source materials (i.e. slide deck or full academic paper) to be used for (RAG) Retrieval-Augmented Generation of the talk concepts and derived statements.
- **Opt-out:** Presenters who believe bingo would distract from their work, or who simply prefer not to participate, can request removal. Their talk remains visible in the schedule, but attendees cannot access a bingo card for it.

This model provides comprehensive coverage by default whilst giving presenters meaningful control. We would like to gather feedback during CHI 2026 to refine these approaches in the future. Presenters will be able to anonymously provide feedback on the system and its use via a form on our website.

2.4 Card Generation and Distribution

Each participant receives a unique bingo card drawn from the session’s statement pool. Our approach attempts to balance the likelihood of each player’s card having bingo success by using a weighted selection of the statements from their assigned probability of appearing (i.e. high, medium, low); we also prompt for a temporal distribution across the talk, and inclusion of any presenter customised statements/key takeaways.

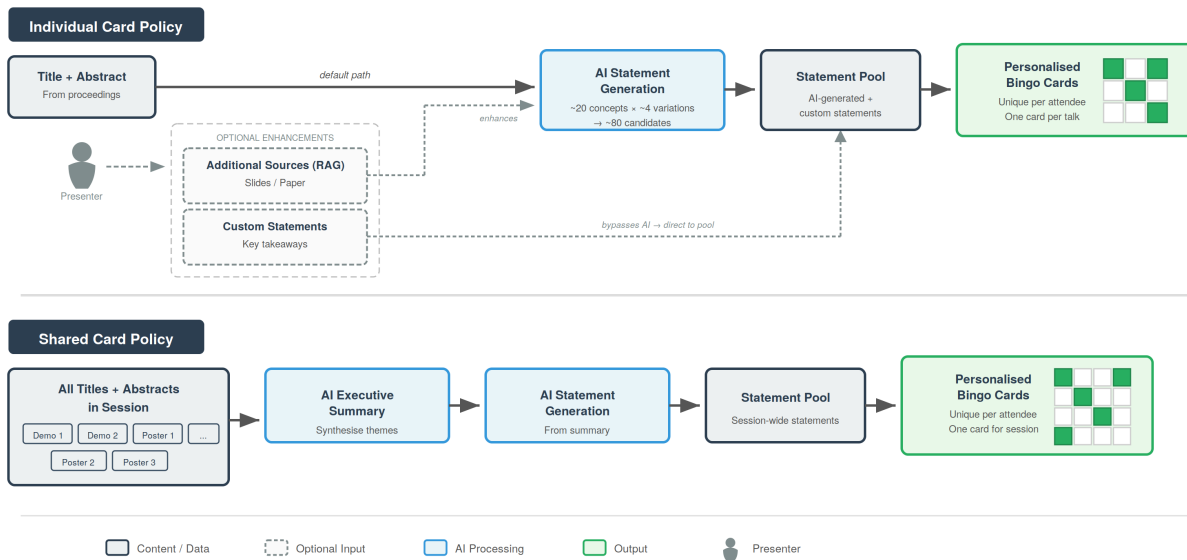


Figure 1: Content generation pipelines for Slide.Bingo's card policy modes (top) Individual (bottom) Shared.

2.5 Badge Scanning and Real-Time Features

To facilitate real-time engagement among CHI conference attendees, the system includes a lightweight registration process. Attendees scan their conference badges using their device's camera. The system extracts name and institution (processed and stored locally in their browser), then we use their first name and institution (for leaderboards, with the option to opt out and not appear on the public leaderboard). Badge scanning verifies that a player is an attendee without requiring user accounts. Only the first name and institution are used publicly, and no badge photographs will be retained. The deployment will comply with GDPR requirements, with a clear privacy notice presented during badge scanning.

WebSocket-based synchronisation creates shared experience through player counts, achievement notifications, and an event-wide leaderboard. Built as a Progressive Web App (PWA) with offline support, gameplay actions are processed opportunistically and queued for synchronisation when connectivity is restored.

3 Interactive Demo at CHI 2026

3.1 Conference-Wide Deployment

Slide.Bingo at CHI 2026 is not merely a booth demonstration; it is a conference-wide deployment. We will populate chi2026.slide.bingo with the complete CHI 2026 programme (excluding papers where authors have indicated that they would like to opt-out), generating bingo content for paper sessions (Individual Card Policy) and demo/poster/interactivity sessions (Shared Card Policy) across the entire event. By embedding Slide.Bingo directly into live sessions across the programme, the deployment aims to encourage attendees to engage actively during presentations and to contribute

lightweight, constructive inputs in real time, rather than solely consuming content passively.

Attendees as participants: Any CHI 2026 attendee can use Slide.Bingo throughout the conference. Upon first use, attendees scan their badge to register. The system then serves as a functional schedule browser while providing bingo gameplay for any session.

Presenters as customizers: CHI 2026 presenters can visit our Interactivity booth to customize their papers' statements, adding custom content for key takeaways or provocations.

3.2 The Interactivity Booth Experience

The booth serves two primary functions: a hands-on demonstration for curious attendees and a customization station for CHI 2026 presenters. Figure ?? shows the proposed booth layout.

Visitor Flow (Attendee Demonstration): A large display continuously loops a pre-recorded video presentation with accompanying Slide.Bingo cards. Visitors scan a QR code displayed at the booth to join the demo session on their own device. They receive a bingo card synchronised to the video and can play along, experiencing the core gameplay loop. This allows multiple participants to participate simultaneously without requiring one-on-one facilitation. Attendees can get a strong sense of how the system operates in 3-5 minutes. Bystanders can view both the video and the live engagement dashboard, which shows real-time engagement metrics.

Visitor Flow (Presenter Customization): Presenters at CHI 2026 can visit the booth to customize the bingo content associated with their CHI 2026 presentations. On the Slide.Bingo website, presenters have the option to prepare their customisations on their personal devices and present a QR code (similar to our previous KaraokAI [3] interactivity) to the Slide.Bingo team to quickly review and action any changes. Team members can also guide them

through the short workflow: uploading slides (optional) and adding custom statements for key takeaways. Changes propagate immediately to the live system.

System Demonstration: For visitors interested in technical details, we can demonstrate the conference setup pipeline, AI statement generation process, and administrative dashboard on a separate screen.

4 Relevance to CHI

Slide.Bingo sits at the intersection of several research themes central to the CHI community. As an interactive system designed to reshape how attendees experience presentations, it embodies HCI's commitment to understanding and improving human engagement with technology. The system draws on established research in active learning, gamification, and social computing, whilst exploring timely questions about the role of generative AI in augmenting rather than replacing human attention.

The CHI conference itself presents an ideal deployment context—hundreds of presentations across diverse formats, and a community open to experimenting with how we experience academic work together. By deploying Slide.Bingo at CHI2026 we will garner feedback from attendees and presenters, on (1) attendees use of Slide.Bingo for sustained active learning in a conference setting and presenters' input on (2) the use of such technologies as part of conferences, (3) the presenter customization and agency workflows we have in place, and (4) their use of Slide.Bingo to promote active learning with the research they are sharing with the conference community. This will feed into our iterative design processes as we further develop Slide.Bingo for deployment at future in-person, hybrid or remote conferences, higher education classrooms and training events.

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References

- [1] David P. Ausubel. 1960. The use of advance organizers in the learning and retention of meaningful verbal material. *Journal of Educational Psychology* 51, 5 (1960), 267–272.
- [2] Charles C. Bonwell and James A. Eison. 1991. *Active Learning: Creating Excitement in the Classroom*. ASHE-ERIC Higher Education Reports.
- [3] Anna R. L. Carter, Dalya Al-Shahrabi, William A. P. Imoukhuede, Lauren Scott, Dan Jackson, Reem Talhouk, Patricia Piedade, and Kyle Montague. 2025. KaraokAI: Exploring the Potential of AI-Generated Karaoke for Academic Conferences and Research Experiences. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*. Association for Computing Machinery, New York, NY, USA, Article 734, 5 pages. doi:10.1145/3706599.3721113
- [4] Diana-Margarita Córdova-Esparza, Julio-Alejandro Romero-González, Karen-Edith Córdova-Esparza, Juan Terven, and Rocío-Edith López-Martínez. 2024. Active learning strategies in computer science education: A systematic review. *Multimodal Technologies and Interaction* 8, 6 (2024), 50.
- [5] Scott Freeman, Sarah L. Eddy, Miles McDonough, Michelle K. Smith, Nnadozie Okoroafor, Hannah Jordt, and Mary Pat Wenderoth. 2014. Active learning increases student performance in science, engineering, and mathematics. *Proceedings of the National Academy of Sciences* 111, 23 (2014), 8410–8415.

arXiv:https://www.pnas.org/doi/pdf/10.1073/pnas.1319030111 doi:10.1073/pnas.1319030111

- [6] Kathrin Hicks et al. 2024. CHI Play Bingo. In *CHI PLAY 2024*. <https://dl.acm.org/doi/fullHtml/10.1145/3665463.3678843>
- [7] Raja Natarajan. 2008. On Attending Conferences. *Computer* 41, 2 (Feb. 2008), 108–107. doi:10.1109/MC.2008.58
- [8] Johanna Pirkker, Maria Riffnaller-Schiefer, and Christian Gütl. 2014. Motivational active learning: engaging university students in computer science education. In *Proceedings of the 2014 conference on Innovation & technology in computer science education*. 297–302.
- [9] Michael Prince. 2004. Does Active Learning Work? A Review of the Research. *Journal of Engineering Education* 93, 3 (2004), 223–231.
- [10] Michael Sailer and Lisa Homner. 2020. The Gamification of Learning: A Meta-analysis. *Educational Psychology Review* 32 (2020), 77–112.